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Set 15: Product Sales Analysis (Unit IV)

Question 1: Bar Chart

Aim

To create a grouped bar chart comparing the monthly sales of each product.

Procedure

1. Load ggplot2 and create the sales data frame.
2. Convert it to long format (Product, Month, Sales).
3. Draw a grouped bar chart with title and axis labels.

Code

```
library(ggplot2); library(gridExtra)
p <- data.frame(ID=1:3, Product=c("Product A","Product B","Product C"),
  Jan=c(2000,1500,1200), Feb=c(2200,1800,1400), Mar=c(2400,1600,1100))
l <- data.frame(Product=rep(p$Product,3),
  Month=factor(rep(c("Jan","Feb","Mar"), each=3),
  levels=c("Jan","Feb","Mar")),
  Sales=c(p$Jan, p$Feb, p$Mar))
b <- ggplot(l, aes(Month, Sales, fill=Product)) +
  geom_col(position="dodge") +
  labs(title="Monthly Sales by Product", x="Month", y="Sales")
print(b)
```

Output

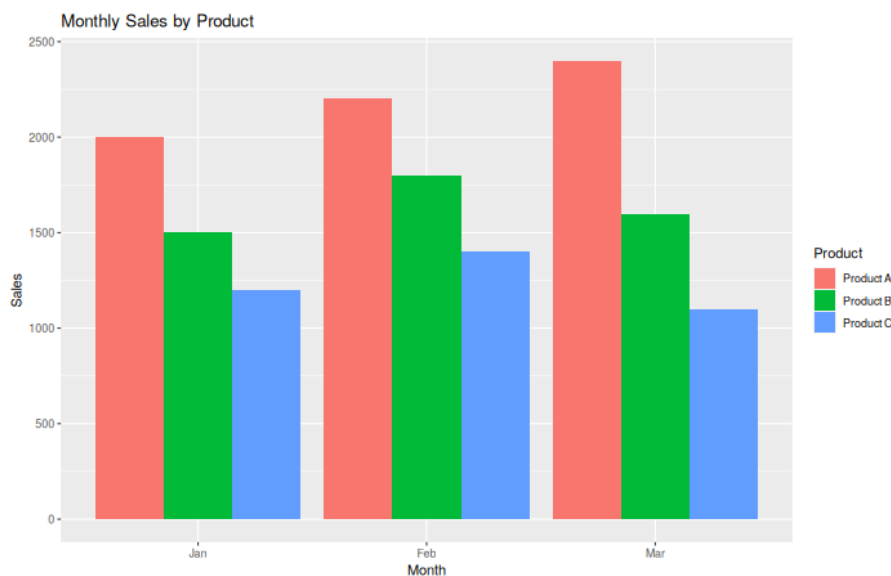


Fig 1: Bar chart – Monthly sales by product

Result

The bar chart was created. Product A has the highest sales in every month.

Question 2: Line Chart

Aim

To show the sales trend of each product from January to March using a line chart.

Procedure

1. Use the long-format data (l).
2. Draw one line per product with points, title and axis labels.

Code

```
n <- ggplot(l, aes(Month, Sales, group=Product, color=Product)) +  
  geom_line() + geom_point() + labs(title="Sales Trend (Jan-Mar)",  
  x="Month", y="Sales")  
print(n)
```

Output

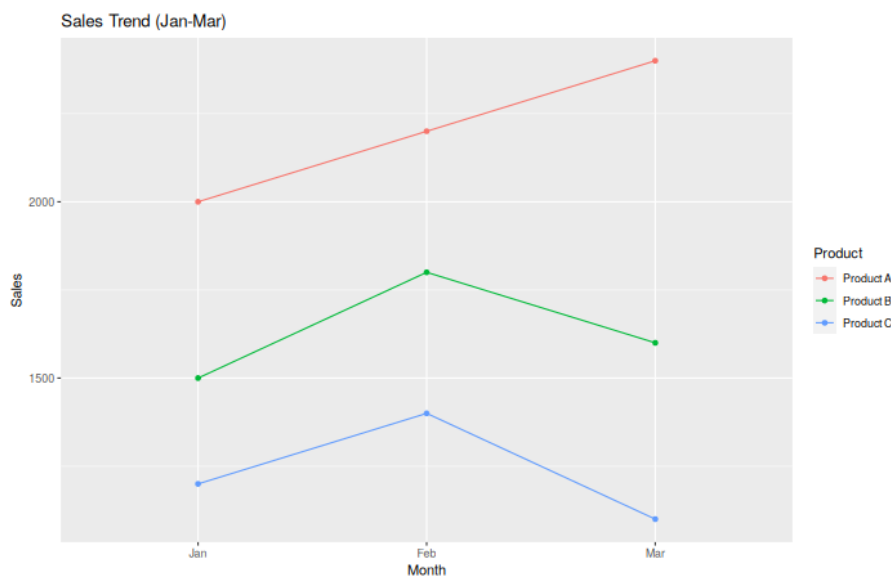


Fig 2: Line chart – Sales trend

Result

The line chart was created. Product A rises every month, while Products B and C fall in March.

Question 3: Table and Dashboard

Aim

To build a sales table with totals and combine it with the charts into one dashboard.

Procedure

1. Add a Total column to the sales data.
2. Print the table and draw it using `tableGrob()`.
3. Combine the bar chart, line chart and table using `grid.arrange()`.

Code

```
p$Total <- p$Jan + p$Feb + p$Mar
print(p)
t <- tableGrob(p, rows=NULL)
grid.arrange(b, n, t, ncol=2)
```

Output

Console output:

ID	Product	Jan	Feb	Mar	Total
1	1 Product A	2000	2200	2400	6600
2	2 Product B	1500	1800	1600	4900
3	3 Product C	1200	1400	1100	3700

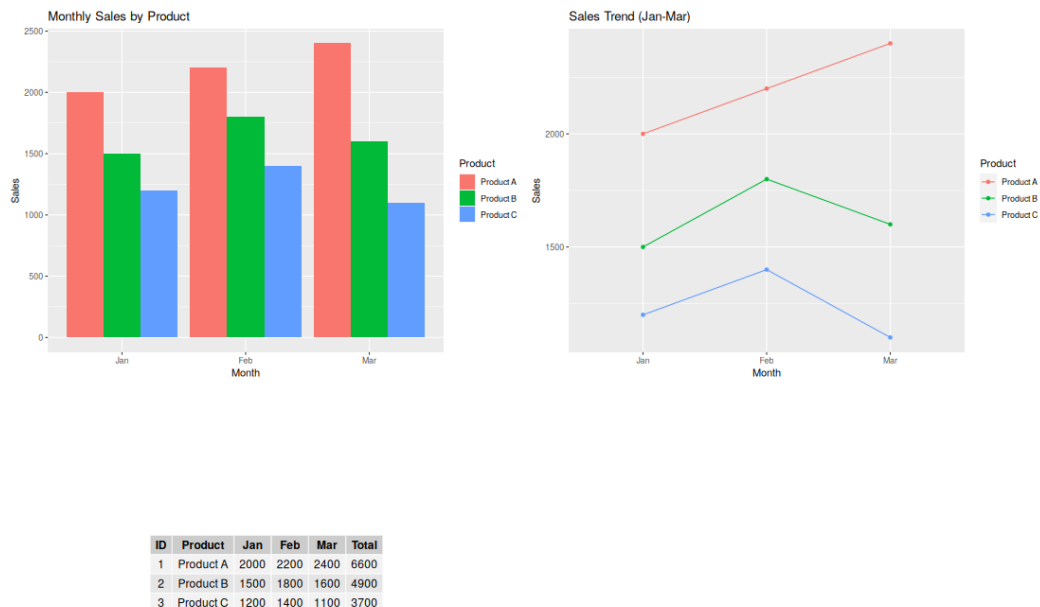


Fig 3: Dashboard (bar + line + table)

Result

The table and dashboard were created. Product A has the highest total sales (6600), then Product B (4900) and Product C (3700).